Urban Expansion and Green Space Loss in Thiruvananthapuram, 2000-2025

Course Project Report, Spatial Big Data

Submitted by	Sebin George	**Submitted to**	Manoj Kumar T K
Roll No.	253309	**Date**	November 2026
Programme	MSc Data Science with Geoinformatics	**Institution**	Digital University Kerala

Abstract. This project measures how much green cover was lost to building in Thiruvananthapuram Corporation (101 wards, 212.7 km²) between 2000 and 2025, and which wards lost the most. For this, dry-season Landsat and Sentinel-2 composites were made in Google Earth Engine and classified with the help of Random Forest, and the large spatial joins were run in Apache Sedona and PostGIS. Built-up land rose from 15.6% to 22.9% of the corporation, whereas green cover fell from 77.7% to 74.2%, and in total 12.2 km² of green land became built-up. Most of this happened in the southern and central wards, close to the roads and to the land which was already built up, and these areas also warmed the most. The rate of loss has slowed, from 0.71 km² a year in 2000-2010 to 0.52 km² a year in 2017-2025.

1. Introduction

Thiruvananthapuram is a very green city, but most of its green is not forest or park. It is homestead land, i.e. houses surrounded by coconut palms, banana and jackfruit trees, and small paddy patches. When someone builds a new house, usually only a part of the plot is cleared, and the trees which are left standing can hide the roof in a satellite image. Hence the growth of the city is hard to see from above.

Over the last 25 years the city has grown along the NH 66 bypass, around Technopark in the north and towards the Vizhinjam port in the south. The aim of this project is to find out how much green cover has been removed by this growth, and in which wards. Green cover keeps the city cooler, and the low-lying paddy fields and wetlands soak up the monsoon water. Green area per person is also a simple way to compare how liveable the different wards are. The numbers are needed at ward level, because the ward councillors and the planning office cannot act on a single figure for the whole city.

The project is also a big data problem, and hence it suits this course. One Sentinel-2 composite of the study area has about 7 million 10 m pixels per band, and the analysis has to join millions of pixels with the ward boundaries, roads and around 203,000 building footprints. So the project had two goals:

1. To measure how much green land turned into built-up land in the 101 wards for 2000, 2010, 2017 and 2025, to find where it is clustered and what makes it more likely, and to rank the wards which need attention first.
2. To do the heavy spatial processing with the tools from the course (Spark with Apache Sedona, PostGIS and FP-Growth in Spark MLlib), and to check what each tool actually adds.

2. Data and methods

2.1 Study area and base data

The 2024 K-SMART ward boundaries of the corporation (local body code C010100, 101 wards) are used. The Census 2011 wards do not match these new boundaries, hence the population is taken from WorldPop grids and added up for each ward. Roads, land-use tags and building footprints are taken from OpenStreetMap. The Overpass API was tried first, but it was not reachable from my network, so the Geofabrik southern-zone extract was downloaded and read directly. Every layer is projected to UTM zone 43N (EPSG:32643), so that the areas and distances come out in metres. The ward file also had surveyor names and phone numbers, and these were removed because they are personal data.

Hand-drawn ward polygons often have geometry problems, such as boundaries which cross themselves or badly formed rings, and these can make overlays and buffers fail or return wrong areas. Hence, after reprojecting, *make_valid()* in GeoPandas was run on all the ward polygons, the ward areas were calculated, and it was checked that there were still exactly 101 wards with unique numbers. Only after this were the wards merged into one corporation outline. The merge left thin slivers in the small gaps between the neighbouring wards, and these were closed with a 10 m buffer out and back in.

2.2 Satellite composites

Kerala is cloudy for most of the year, hence only dry-season images (November to April) are used, and the median of all the clear observations is taken in Google Earth Engine. For 2000 and 2010, Landsat 5 and 7 had to be combined over two dry seasons, because Landsat 5 alone had no usable scenes for 2000. Landsat 8 is used for 2017, Landsat 8 and 9 for 2025, and Sentinel-2 with Cloud Score+ masking ($cs_cdf \geq 0.60$) for 2019 and 2025. Every composite covers at least 98.6% of the area, and the median number of clear observations per pixel is between 7 and 29. NDVI, NDBI and MNDWI are added to the six reflectance bands.

2.3 Land-cover classification

Five classes are mapped, i.e. built-up, tree cover, other vegetation, water and bare/sand. Getting good training labels was the hardest part of the whole project. The plan was to take the labels from global land-cover products, but these disagree a lot in this city. Dynamic World calls about 80% of the corporation built-up, whereas ESA WorldCover calls most of it trees. Both are looking at the same coconut-covered homesteads and reading them differently. Hence, in the end, the Sentinel-2 training points were taken only where WorldCover and Dynamic World agree, with an extra check against GHSL built-up density. For Landsat, my Sentinel-2 2025 map aggregated to 30 m is used as the labels, so that both the resolutions use the same class definitions.

The classifier used is a Random Forest with 200 trees, where each split looks at $\sqrt{p}$ of the p input features. For each image, about 7,000 training points are used and about 3,000 are kept for testing. The points are split by spatial blocks, because otherwise a test point can sit right next to a training point and the accuracy looks better than it really is. Land surface temperature was also considered as a feature, but Landsat LST is missing over roughly 20% of the area in every year, hence it was left out.

2.4 Making the years consistent

The first try was to classify each year on its own and compare the maps. This gave almost no growth in built-up land (20.9% to 22.9%), which clearly did not match GHSL (13.2% to 21.3%). The problem is that each year's map has its own classification errors, and when two maps are compared, these errors get mixed up with the real change. Hence the work was done backwards instead, i.e. starting from the 2025 map and stepping back one epoch at a time. A pixel keeps its later class unless its spectral values changed by more than k standard deviations between the two composites; only then does it take that year's own class. k was tested from 0 to 1.5 and k = 0.25 was chosen, because its built-up trend fitted GHSL the best

(squared error 0.011). At 30 m the model could not reliably separate tree cover from other vegetation, hence in the results these two are added together as total green.

2.5 Big data processing

Every pixel in the study area (the corporation plus a 2 km buffer) is turned into a point which carries its class history, i.e. 451,000 points at 30 m and 4.04 million at 10 m. These points are then processed in two systems.

The first is Apache Sedona on Spark, running in local mode on my laptop. The pixels are joined to the wards in four ways: plain Spark with no spatial index (nested loop), a Sedona broadcast join with an R-tree, and Sedona partitioned joins with KDB-tree and Quadtree partitioning. These are compared with GeoPandas at 0.1, 0.5, 1 and 4 million points. Sedona is also used to find the distance from every green pixel to the nearest main road, and to sum the change into 2,082 H3 hexagons (resolution 9).

The second is PostgreSQL 13 with PostGIS. Here the same joins, a 1 km range query around Technopark and a nearest-road (KNN) query are run, each with and without a GiST index. The pixel table is also split into 30 range partitions in order to test partition pruning, and the query plans are saved with EXPLAIN ANALYZE. To check the joins, the ward totals from Sedona, PostGIS and the raster pipeline are compared; they differed by at most 0.46 percentage points.

2.6 Analysis and decision support

In order to find the hotspots, Global Moran's I and Getis-Ord Gi* are used on the hexagons (6 nearest neighbours, 999 permutations, false discovery rate correction). DBSCAN (eps 100 m, 12 pixels) is also run on the converted pixels to find compact building sites.

For the association rules, every pixel which was green in 2000 becomes one transaction made of simple attributes, i.e. how built-up its surroundings were in 2000, distance to the centre, roads and Technopark, elevation, slope, low-lying or not, ward density, vegetation type and OSM land use. FP-Growth is run in Spark MLlib (min. support 0.003, min. confidence 0.10), and the rules which end in "built-up" or "stayed green" are kept.

The case studies look at heat (MODIS at 1 km, and Landsat LST), flooding (low-lying land with MERIT Hydro HAND ≤ 5 m, and the 2018 flood mapped from Sentinel-1) and green area per person. For the ward priority index, seven ward indicators are taken (long-term and recent green loss, share in hotspots, low-lying green converted, LST in 2025, green per person and density), min-max scaled and averaged. Three other weightings were also tried to see whether the ranking holds. Finally, all the results are put into a Streamlit dashboard, so that people who are not GIS specialists can explore them.

3. Results

3.1 Classification accuracy

Table 1. Random Forest accuracy on the held-out test points.

Composite	Train / test	Overall acc.	Kappa	F1 built-up	F1 tree	F1 other veg.	F1 water
Sentinel-2 2019	6,857 / 3,143	0.959	0.948	0.926	0.966	0.924	1.000
Sentinel-2 2025	6,786 / 3,214	0.958	0.947	0.962	0.933	0.944	1.000
Landsat 2000	6,989 / 3,011	0.709	0.635	0.722	0.692	0.428	0.984
Landsat 2010	6,741 / 3,259	0.757	0.696	0.699	0.738	0.509	0.960
Landsat 2017	6,990 / 3,010	0.823	0.778	0.724	0.814	0.592	0.994
Landsat 2025	7,154 / 2,846	0.819	0.773	0.771	0.817	0.609	0.993

The test labels come from the same reference products as the training labels, hence these figures measure the agreement with those products. No field data was available to test against.

The Sentinel-2 maps reached about 96% accuracy. Landsat is lower (71-82%) and gets worse for the older images, mostly because other vegetation is confused with tree cover (F1 only 0.43-0.61). This is the reason why the two are merged into one green class. Water was classified well in every year, and built-up land reasonably well (F1 0.70-0.96).

3.2 How much green was lost

Table 2. Green-cover change in the corporation (backdated Landsat series, km²).

Period	Green at start	Green at end	Green lost	of which to built-up	Green gained	Net change	Loss per year
2000-2010	165.15	161.50	7.10	5.26	3.44	−3.65 (−2.2%)	0.71
2010-2017	161.50	159.35	4.37	2.94	2.22	−2.15 (−1.3%)	0.62
2017-2025	159.35	157.72	4.19	3.43	2.56	−1.63 (−1.0%)	0.52
2000-2025	165.15	157.72	13.34	12.17	5.91	−7.43 (−4.5%)	0.53

Built-up land grew from 33.2 km² (15.6%) in 2000 to 48.8 km² (22.9%) in 2025, i.e. an increase of 47%, whereas green cover went down from 77.7% to 74.2%. Of the 13.3 km² of green which was lost, 12.2 km² (91%) became built-up. The rest became water or bare land. About 5.9 km² of new green offset a part of the loss, and most of it is on the bare and sandy land along the north-west coast. The yearly loss became smaller in each period, from 0.71 to 0.52 km².

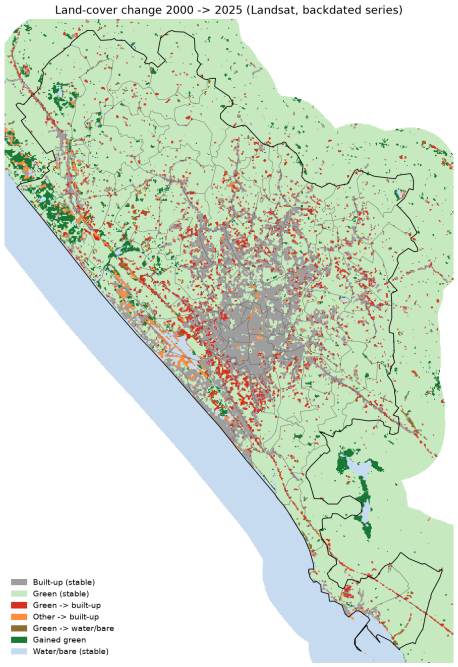


Figure 1. Land-cover change 2000-2025. Red is the green land which became built-up. The conversion spreads out from the dense core (grey) along the coastal highway to the south and along the main roads to the north and east. Green gains (dark green) are mostly on the north-west coast.

3.3 Big data processing performance

Table 3. Point-in-polygon join of pixels to the 101 wards, run time in seconds.

Points	Spark nested loop	Sedona broadcast R-tree	Sedona Quadtree	Sedona KDB-tree	GeoPandas (STRtree)
0.10 M	4.4	0.31	0.67	1.13	0.04
0.50 M	19.6	0.37	1.93	3.03	0.31
1.00 M	40.2	0.71	3.83	6.56	0.76
4.04 M	192.0	3.00	17.56	34.00	1.63

The spatial index made the biggest difference. At 4 million points, the Sedona broadcast R-tree join took 3 seconds, whereas plain Spark took 192, i.e. about 64 times slower. Since there are only 101 ward polygons, sending the whole ward layer to every worker worked better than partitioning both the sides. Between the two partitioners, Quadtree gave more even partitions than KDB-tree (coefficient of variation of points per partition 0.44 against 1.79) and ran about twice as fast. Spark's default hash partitioning makes every partition the same size, but the points in each partition are spread over 99.9% of the city's extent, hence it gives no spatial pruning at all. What surprised me was that GeoPandas was still the fastest on my laptop. Hence, at this data size, the reason to use Sedona is that the same code can scale to a cluster and to a larger area.

Table 4. PostGIS queries with and without a GiST index (seconds).

Query	No index	GiST index	Speed-up
451 k pixels to wards (point-in-polygon)	4.29	0.53	8×
203 k building centroids to wards	20.60	0.87	24×
Range query: converted pixels within 1 km of Technopark	0.073	0.009	8×
KNN: nearest main road for sampled green pixels	14.54	0.13	114×
Partitioned table: 1 of 30 partitions vs all 30	0.076	0.010	8×

In PostGIS, the GiST index replaced the sequential scans with index scans. The KNN query gained the most (114×), because the <-> operator can walk the index in the order of distance. With partitioning, PostGIS read only 1 of the 30 partitions when the query filtered on the partition key. PostGIS and Sedona both gave the same green to built-up total of 12.17 km².

3.4 Where the loss is concentrated

The conversion is strongly clustered (Moran's I = 0.485 for hexagons and 0.343 for wards, p = 0.001). Gi* found 262 hot hexagons at the 95% level (115 after FDR correction), which hold 4.55 km² of conversion. DBSCAN grouped 57.5% of all the converted pixels into 250 compact clusters. The largest ones are in Sreevaraham (0.44 km²), Kalippankulam (0.40 km²) and Chakkai (0.28 km²), all south of the core near the bypass and the airport. The cold spots, where the green was kept, are along the northern, eastern and southern edges of the corporation.

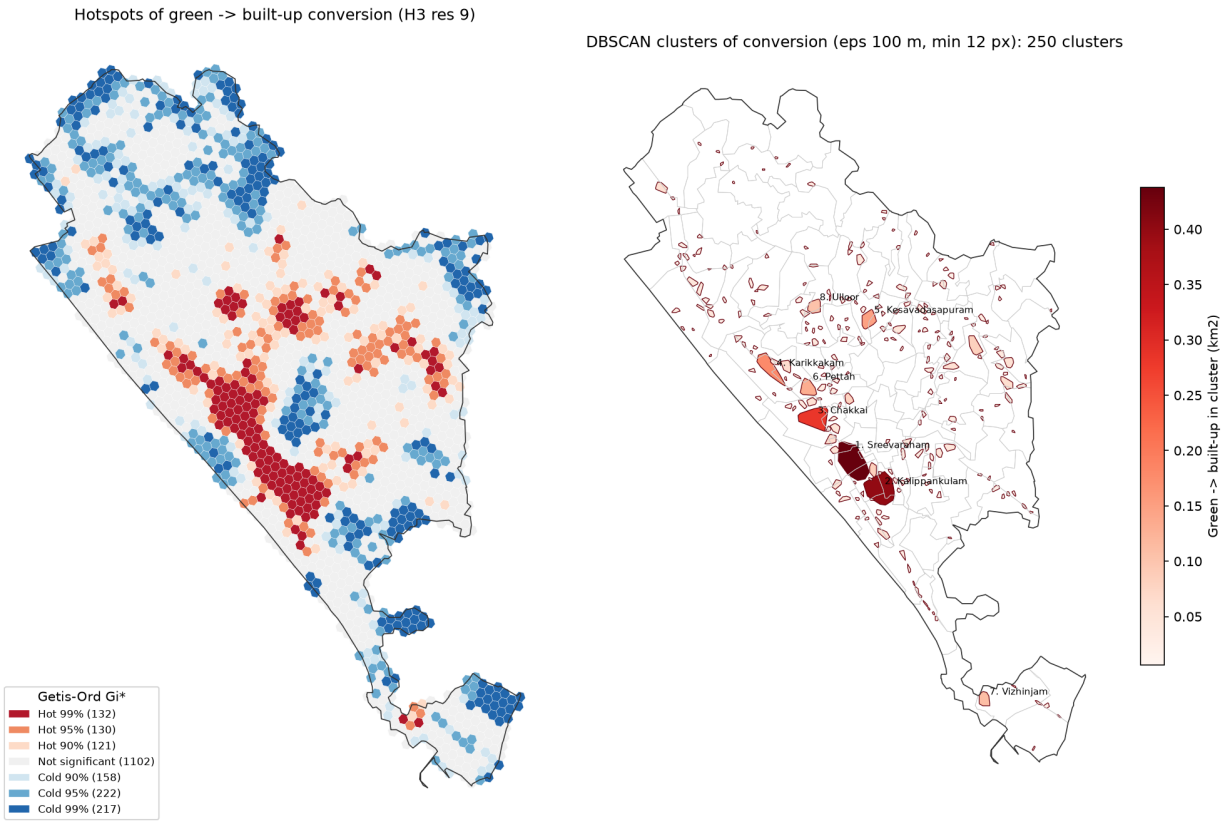


Figure 2. Left: Getis-Ord Gi* hot and cold spots of green to built-up conversion on H3 hexagons. Right: DBSCAN clusters of the converted pixels, with the eight largest labelled by ward.

3.5 What makes conversion more likely

Distance to the roads had a strong effect. Of the green land within 100 m of a road, 12.4% was converted, whereas only 2.3% was converted at 1-2 km from a road. Overall, 61% of all the conversion happened within 250 m of a road.

Table 5. Strongest FP-Growth rules (182,192 transactions; the base conversion rate is 7.42%).

Condition (green pixel in 2000)	Outcome	Area (km²)	Confidence	Lift
high built-up around it, <3 km from centre, flat	built-up	1.35	39.4%	5.31
high built-up around it, elevation <10 m	built-up	1.67	35.4%	4.76
high built-up around it, flat, high ward density	built-up	2.21	35.2%	4.75
high built-up around it, elevation 10-30 m	built-up	7.78	29.8%	4.02
OSM land use = forest	stayed green	2.46	99.3%	1.07
OSM land use = agriculture	stayed green	2.69	99.3%	1.07
>10 km from centre	stayed green	34.51	97.1%	1.05

FP-Growth found 146 rules for conversion and 17 for staying green. Most of the strong ones describe infill, i.e. 27% of the green land which was already surrounded by built-up land in 2000 was converted, compared with 3.5% where the surroundings were green. Flat, low land near the centre was converted the most often. Land tagged in OSM as forest, agriculture or wetland almost always stayed green.

3.6 Heat, flooding and green per person

Greener wards are found to be cooler. Ward green share against MODIS LST 2025 gives Spearman $\rho = -0.77$ ($p < 10^{-20}$). In 2025, Landsat LST averaged 38.7 °C on built-up land, whereas it was 34.6 °C under tree cover. Pixels which changed from green to built-up warmed by 3.0 °C between 2000 and 2025, against 1.5 °C for the pixels which stayed green, i.e. about 1.6 °C of extra warming.

Low-lying land was converted slightly more often. Of the low-lying green land (HAND ≤ 5 m), 7.8% was converted (2.95 km²), against 7.2% on higher ground. The 2018 flood extent which I mapped inside the corporation was only 0.32 km², and almost no conversion happened inside it. Hence low-lying land, and not the 2018 flood extent, is used to discuss the flood risk.

Green area per resident fell from 180 m² in 2000 to 171 m² in 2010 (WorldPop 2000-2020 release), and from 139 m² in 2017 to 132 m² in 2025 (WorldPop R2025A). The number of wards below 50 m² per person went up from 17 to 27. The two WorldPop releases cannot be compared with each other, hence the drop between 2010 and 2017 is not interpreted.

3.7 Ward priority

The top five wards on the priority index are Sreevaraham (95/100), Perunthanni (88), Kamaleswaram (86), Pettah (77) and Kalippankulam (74), and all of them are in the southern part of the core. Changing the weights hardly moves the ranking. For every weighting, the Spearman correlation with the equal-weight ranking is at least 0.967, and 16 to 19 of the top 20 wards stay in the top 20.

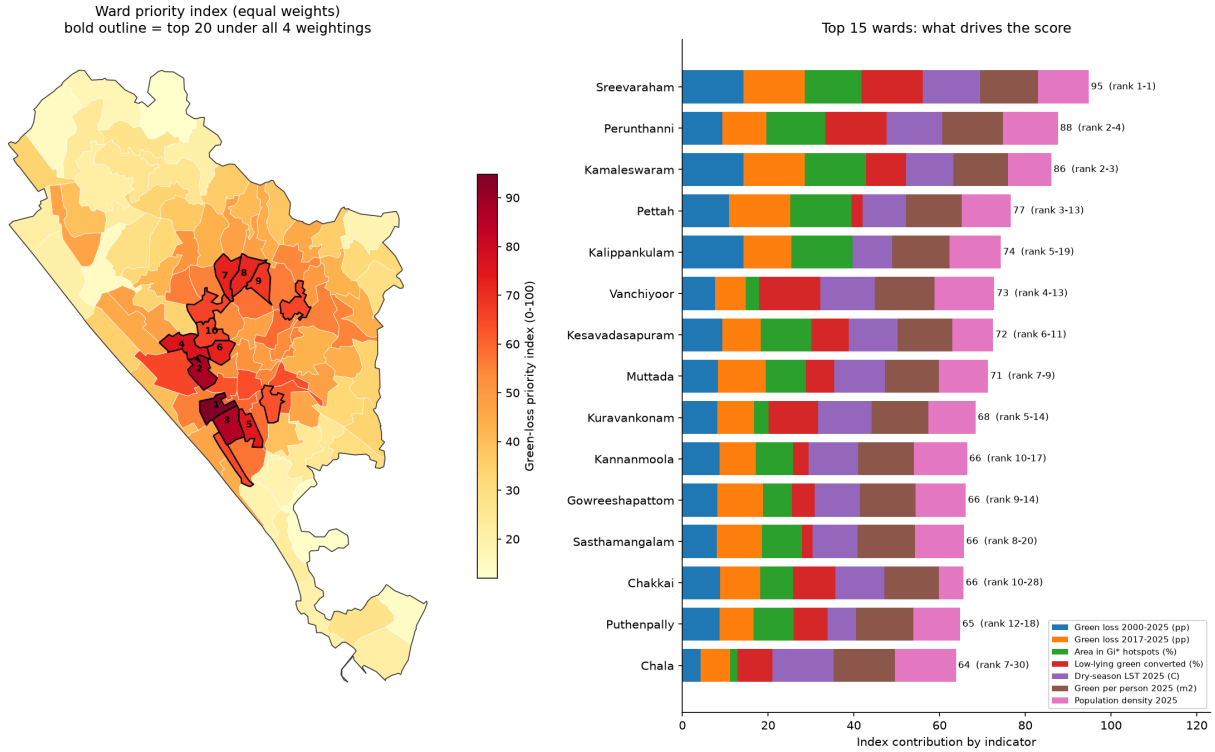


Figure 3. Ward priority index (equal weights). Left: map, with a bold outline for the wards which are in the top 20 under all four weightings. Right: what each indicator adds for the 15 highest-scoring wards, along with the range of ranks across weightings.

4. Discussion

4.1 What the results mean

Thiruvananthapuram lost about 4.5% of its green cover in 25 years. This sounds small next to the 47% growth in built-up land, but the city started out very green, and a lot of the new building happened inside the existing homesteads. The loss is also uneven. Most of it is in a belt south and west of the old city (Sreevaraham, Kalippankulam, Kamaleswaram, Chakkai, Pettah) and along the highways, where some wards lost 18 to 30 percentage points of their green cover. The association rules and the road-distance results point to the same process, i.e. new building fills the gaps next to the existing buildings and roads, and very little of it appears in open land away from them. The slower loss after 2010 may be because less green is left near the core, but my data cannot confirm this or rule out other causes.

Heat is where this loss is the easiest to feel. Land which was built on warmed 1.6 °C more than the land which stayed green, and the priority wards are among the hottest and most crowded in the city. Hence keeping the green which is left there should help directly with the heat. Land recorded as agriculture, forest or wetland in OSM was almost never built on. This suggests that land with a recognised use is better protected, although an OSM tag is not the same thing as an official land-use designation.

4.2 What I learned about the big data tools

The main thing I learned is that a spatial index matters more than the choice of engine. An R-tree in Sedona or a GiST index in PostGIS gave speed-ups of 8 to 114 times, whereas Spark without spatial support was slow. Partitioning also has to follow the data, i.e. hash partitioning balances the load but ignores the location, whereas the Quadtree partitions adapted to the dense core. For one city with 4 million points, a single machine is still enough. Sedona would become necessary if the work is extended to the whole state, to 10 m pixels for every year, or to building-level analysis. PostGIS was the easiest for the repeated ward and distance queries, and Spark was the natural place to run FP-Growth on 182,000 transactions. Running the same joins in both the systems was also a good way to check that my numbers were right.

4.3 Limitations

- Accuracy was checked only against labels from global products, because no field data was available. Landsat 2000 has the lowest accuracy (71%), and its 6.7% bare/sand share looks too high. This is what produces the apparent green gains in north-western coastal wards such as Pallithura.
- At 30 m, a homestead with a new house under coconut trees often still looks green, hence the conversion is probably underestimated, especially before 2017.
- The backdated series depends on the choice of k and on GHSL as the reference. With k between 0 and 0.5, the total drop in green share ranges from 4.3 to 2.7 percentage points (3.5 at k = 0.25).
- MODIS LST is coarse (1 km), Landsat LST has gaps, the two WorldPop releases cannot be compared across 2010-2017, and OSM is not equally complete everywhere.
- Sedona was run only in local mode on one machine, i.e. a real multi-node cluster was not tested.

4.4 Conclusion and future work

Between 2000 and 2025, Thiruvananthapuram lost 12.2 km² of green land to building, mostly through infill near the roads and the existing built-up areas in the southern and central wards. These same wards are the hottest and have the least green per person. Since the priority ranking barely changes with the weights, it gives the corporation a reasonable shortlist of wards where green space should be protected and trees should be planted. If this work is continued, I would like to collect field validation points, use building footprints from different years to catch the houses hidden under trees, run the workflow on a real Spark cluster for the surrounding panchayats, and update the dashboard every dry season.

References

1. Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5-32.
2. Brown, C. F. et al. (2022). Dynamic World, near real-time global 10 m land use land cover mapping. *Scientific Data*, 9, 251.
3. Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200. Zenodo.
4. Pesaresi, M. et al. (2023). GHS-BUILT-S R2023A: GHS built-up surface grid. European Commission, Joint Research Centre.
5. Gorelick, N. et al. (2017). Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18-27.
6. Yu, J., Zhang, Z., Sarwat, M. (2019). Spatial data management in Apache Spark: the GeoSpark perspective and beyond. *GeoInformatica*, 23, 37-78.
7. Han, J., Pei, J., Yin, Y. (2000). Mining frequent patterns without candidate generation. *Proc. ACM SIGMOD*, 1-12.
8. Getis, A., Ord, J. K. (1992). The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3), 189-206.
9. Ester, M., Kriegel, H.-P., Sander, J., Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. *Proc. KDD*, 226-231.
10. Yamazaki, D. et al. (2019). MERIT Hydro: a high-resolution global hydrography map. *Water Resources Research*, 55, 5053-5073.
11. WorldPop (2018, 2025). Global high-resolution population denominators. University of Southampton.
12. Kerala Local Self Government Department (2024). K-SMART ward delimitation boundaries.